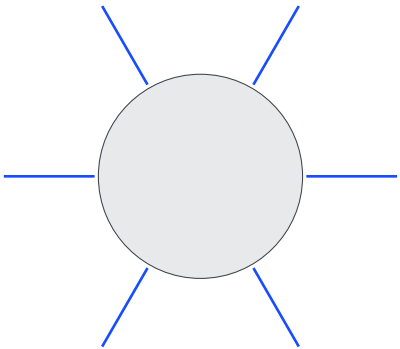
圆柱外任意阶背景模态：同阶负幂镜像模态保证圆周接地



m 阶角向正负瓣

$$\varphi = -K\left(\rho^m - \frac{R^{2m}}{\rho^m}\right)\cos m\phi$$

$\rho = R$ 时两项相消；$\rho \to \infty$ 时恢复给定背景势。

$$\sigma(\phi) = 2\varepsilon_0 K m R^{m-1}\cos m\phi$$